JUDGES’ LIBRARY / DOSSIER 04

Safety, impact & next steps.

A credible pathway from a promising interface to a rigorously evaluated future system.

Core principle: a review flag is a prompt to examine context, not a diagnosis; no flag is not a safety clearance. Severe or sudden symptoms and approved mission procedures take precedence.

What exists today

A research prototype with a synthetic story, optional local input, deterministic personal comparisons, a schematic body-system navigator, separately labeled NASA research, NOAA context and a human-readable handover. It is built to make assumptions visible and to work after caching, rather than to make a clinical decision on the user’s behalf.

What does not exist

Not implemented / not established	Why the distinction matters
Clinical validation or flight qualification	No sensitivity, specificity, outcome benefit, regulatory approval or NASA endorsement is claimed.
Live medical devices or astronaut telemetry	The demonstration uses generated readings; local entries are user-supplied.
Encrypted records, accounts or remote sync	Local browser storage and exports are not a secure clinical record system.
Automatic treatment or causal inference	Co-occurring changes do not prove a common cause; routine prompts are not prescriptions.

The atlas is a schematic navigation aid. It does not scan an organ or establish its condition. NOAA Kp does not measure individual space-radiation exposure.

02 / RISK CONTROLS

Safety and privacy by design.

Controls currently implemented

Explainability: source, units, personal window, coverage and a visible rule can be inspected. **Data quality:** invalid and missing measurements do not become normal values; invalid CSV import is atomic. **Separation:** synthetic, local and historical-research records remain distinct. **Provenance:** OSD-575 source bytes are checked against a manifest where supported.

Privacy realities

Readings, notes, preferences and action states are stored in browser localStorage. The application has no health-record submission endpoint. Browser storage can be read by same-origin scripts, compromised by extensions or access to the device, denied or erased. Exports are ordinary unencrypted files. External sites and hosting providers have their own data practices.

Use non-identifying data for demonstrations. Production use would require an accountable operator, access controls, encryption, incident response, retention and recovery procedures, and an appropriate legal and regulatory review.

Questions a judge may ask

Question	Grounded answer
Is this a medical device?	No. It is a non-diagnostic prototype and has not undergone a medical or regulatory assessment.
Does NASA data validate the alert?	No. OSD-575 is historical research displayed separately; it does not determine personal thresholds.
Can it run offline?	After an initial successful cache, the app and bundled resources can work without network access; storage restrictions and eviction remain possible.
Where does a handover go?	An export is created for the user; it is not automatically transmitted to a clinician.

03 / FUTURE WORK

Impact requires evidence.

A practical evaluation roadmap

Phase	Work required	Evidence of progress
1 Understand	Consult crew, flight surgeons and human-factors experts; identify tasks and hazards.	Documented user needs, task analysis and hazard register.
2 Protect	Threat-model the workflow; design access, encryption, audit and recovery.	Independent security review and tested recovery procedures.
3 Measure	Validate sensors, input quality and data provenance; specify acceptable use.	Measurement protocols and device-integration testing.
4 Evaluate	Prospectively assess usability, safety and algorithm behavior in appropriate populations.	Predefined outcomes, documented false alarms/missed changes and independent clinical review.
5 Govern	Assess regulation, ownership, monitoring and operational accountability.	Explicit go/no-go decisions before any clinical or flight use.

These are proposed next steps, not completed integrations or established outcomes. Potential Earth applications, such as supporting longitudinal self-observation in remote settings, would also require separate validation in those contexts.

Transparent development

AI-assisted tools were used in website updates, documentation and presentation production. The team must review the complete submission and disclose all AI tools and uses that actually apply. A polished presentation is not a substitute for scientific or operational review.

Sources & further reading

- NASA Human Research Program hazards
- NASA standard measures
- Project security and limitations
- Methodology and scientific boundaries
- Official participant FAQs